

2.2: Graphs of Linear Functions

Definition. (Parallel and Perpendicular lines)

Let L_1 and L_2 be lines with slopes m_1 and m_2 respectively. If L_1 and L_2 are *parallel*, then

$$m_1 = m_2.$$

If L_1 and L_2 are *perpendicular*, then

$$m_1 = -\frac{1}{m_2}.$$

Example.

Find the line *parallel* to $y = \frac{3}{2}x + 1$ that passes through the point $(-4, 10)$.

Find the line *perpendicular* to $y = \frac{3}{2}x + 1$ that passes through the point $(-3, 4)$.

Definition.

The point(s) where a graph intersects the axes are called intercepts.

The **x -intercepts** are the points where a function intersects the x -axis.

The **y -intercept** is the point where a function intersects the y -axis.

- To solve for the y -intercept:
 - Set $x = 0$,
 - Solve for y .
- To solve for the x -intercept:
 - Set $y = 0$,
 - Solve for x .

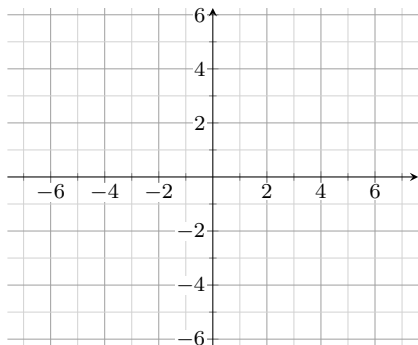
Example. Find the x and y intercepts of the following:

$$y = \frac{4}{5}x - 8$$

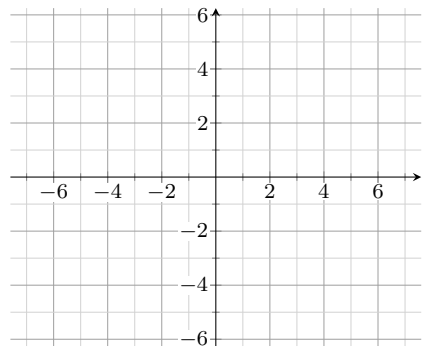
$$3x + 2y = 18$$

Example. Graph the following:

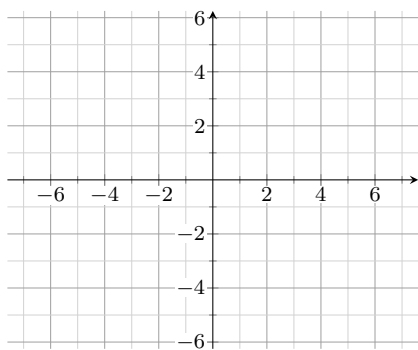
$$y = -\frac{3}{2}x + 3$$



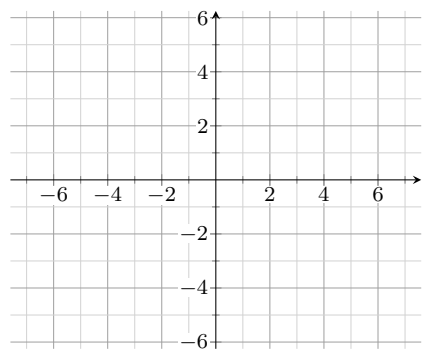
The line through $(-4, 2)$ and $(1, 4)$



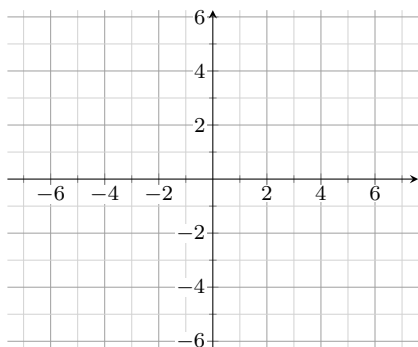
The horizontal line through $(2, 3)$



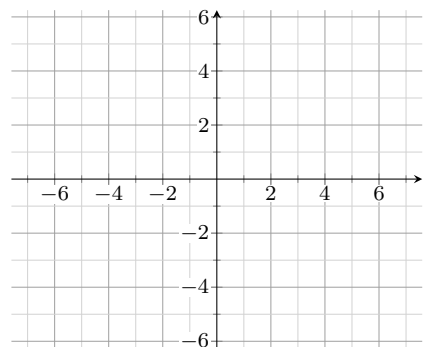
The vertical line through $(-1, 5)$



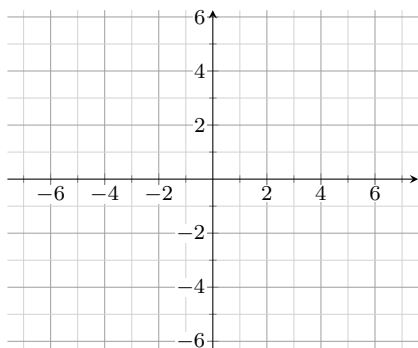
$$8x - 10y = 40$$



$$y = -4x - 5$$



Through $(1, 1)$, parallel to $y = 2x - 4$



Through $(3, -2)$, perp. to $y = \frac{1}{3}x - 1$

